

Neural Collapse by Design: Learning Class Prototypes on the Hypersphere

Koromilas et al. - ICML 2026 [1]

(Edoardo Canti - workshop presentation - Deep Learning Applications 25/26)

Neural Collapse

Neural Collapse is a phenomenon discovered by Papayan et al, 2020 in which:

NC1: within class features collapse to their means $\Sigma \rightarrow 0$

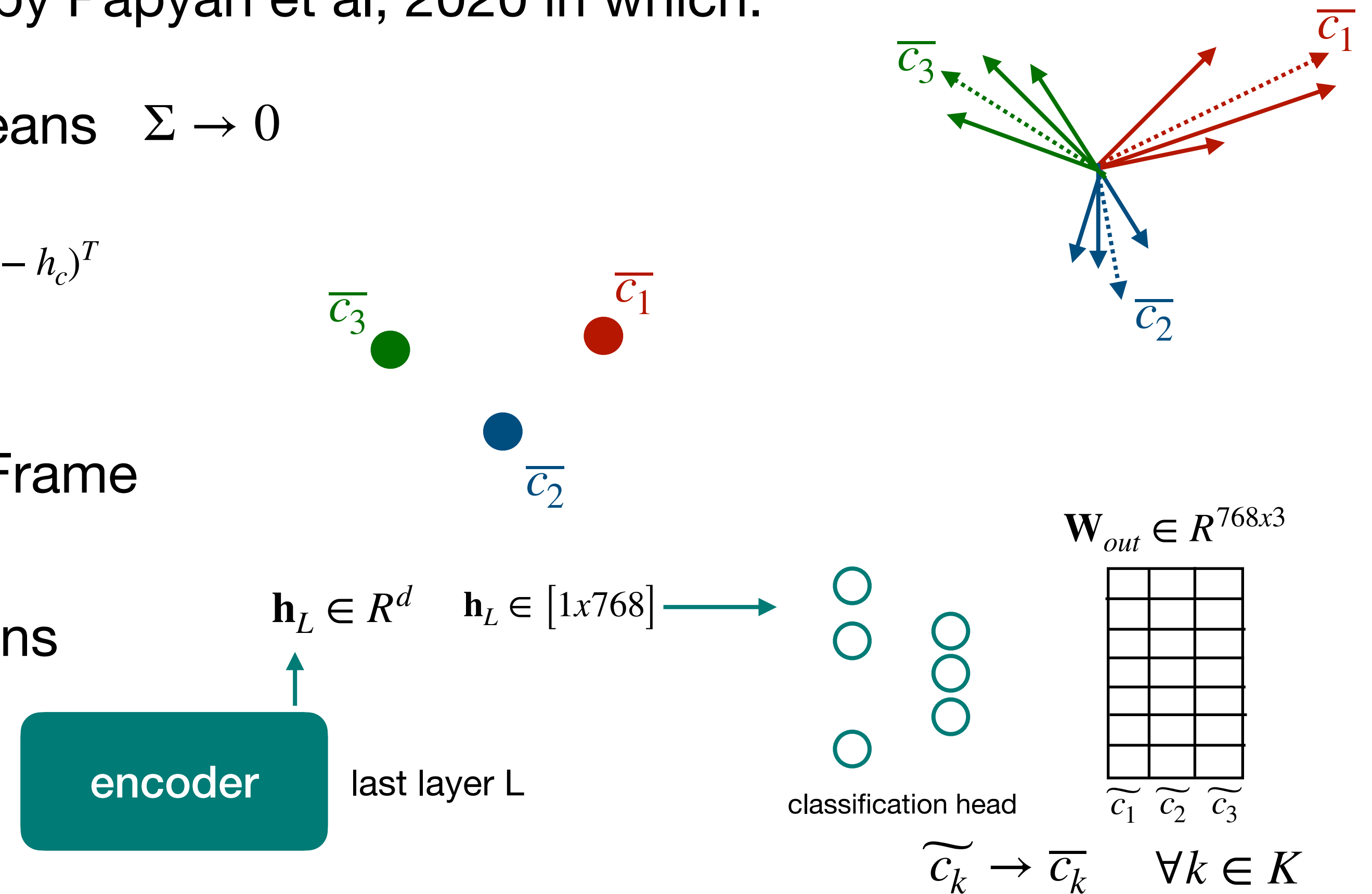
$$\Sigma := \text{AVG}_{i,c} \{ (h_i - h_c)(h_i - h_c)^T \} \Rightarrow \sigma^2 \equiv (h_i - h_c)(h_i - h_c)^T$$

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (h_i - h_c)^2$$

NC2: class means form an Equiangular Tight Frame

NC3: classifier weights align with classes means

NC4: bias collapse



Is Neural Collapse desirable?

Classification problems can be solved via a features extraction pipeline prepended to a classification head. If Neural Collapse arises the “decision rule” reduces to **Nearest Mean Classifier**.

Reaching NC with Cross Entropy is difficult

Learning with Cross Entropy provides a classifier with loss: $\mathcal{L}_{CE(\mathbf{Z}, \mathbf{W})} = \frac{1}{M} \sum_{i=1}^M -\log\left(\frac{e^{\mathbf{z}_i^T \mathbf{w}_{y_i}}}{\sum_{j=1}^K e^{\mathbf{z}_i^T \mathbf{w}_j}}\right)$

If we consider a single instance of the set and isolate the log:

$$\begin{aligned} -\log\left(\frac{e^{\mathbf{z}_i^T \mathbf{w}_{y_i}}}{\sum_{j=1}^K e^{\mathbf{z}_i^T \mathbf{w}_j}}\right) &= -\left[\log\left(e^{\mathbf{z}_i^T \mathbf{w}_{y_i}}\right) - \log\left(\sum_{j=1}^K e^{\mathbf{z}_i^T \mathbf{w}_j}\right)\right] = -\log\left(e^{\mathbf{z}_i^T \mathbf{w}_{y_i}}\right) + \log\left(\sum_{j=1}^K e^{\mathbf{z}_i^T \mathbf{w}_j}\right) \\ &= -\log\left(e^{\mathbf{z}_i^T \mathbf{w}_{y_i}}\right) + \log\left(\sum_{j:y_i=y_j} e^{\mathbf{z}_i^T \mathbf{w}_j} + \sum_{j:y_i \neq y_j} e^{\mathbf{z}_i^T \mathbf{w}_j}\right) = -\log\left(e^{\mathbf{z}_i^T \mathbf{w}_{y_i}}\right) + \log\left(e^{\mathbf{z}_i^T \mathbf{w}_{y_i}} + \sum_{j:y_i \neq y_j} e^{\mathbf{z}_i^T \mathbf{w}_j}\right). \end{aligned}$$

Minimizing this loss function corresponds to maximizing the first term and minimizing the second i.e:

$$\log(e^{\mathbf{z}_i^T \mathbf{w}_{y_i}}) \rightarrow +\infty \implies \|\mathbf{z}_i\| \|\mathbf{w}_{y_i}\| \cos(\theta) \rightarrow +\infty$$

$$\|\mathbf{z}_i\| \geq 0 \quad \|\mathbf{w}_{y_i}\| \geq 0 \implies \cos(\theta) \in (0, 1]$$

Neural Collapse expects a within class features collapse that is not forced, the model could just change vectors amplitudes in order to optimize its goal

Reaching NC with SCL is difficult

In Supervised Contrastive Learning [2](Khosla et al, 2021) the proposed framework is:

1. Data Augmentation module: for each input generate two random augmentation of the input itself. These two copies are named *views*.
2. An Encoder network processes both views independently and then each representation is *normalized (i.e: stands on the hypersphere)*.
3. *A Projection head is then used in order to map the encoder (normalized output) into an alternative feature space of different dimension, the output of this projection layer is then normalized again*

Khosla's methodology then discards the projection head and trains a linear classifier on top freezing the backbone and using **Cross Entropy Loss (Linear Probing phase)**.

We can conclude that, by Khosla's construction, SCL suffers of the same problem of CE.

$$\mathcal{L}_{SCL} = \frac{1}{2M} \sum_{i=1}^{2M} - \frac{1}{|C(i)|} \sum_{l \in C(i)} \log \left(\frac{e^{\frac{\mathbf{a}_i^T \mathbf{a}_l}{\tau}}}{\sum_{j \neq i, j=1}^{2M} e^{\frac{\mathbf{a}_i^T \mathbf{a}_a}{\tau}}} \right)$$

Revisiting Cross Entropy 1/2

NormFace CE projects representations onto the hypersphere in order to **force** the network to focus on θ geometry (avoiding amplitudes corrections).

$$\text{Let } \mathbf{u}_i^T = \frac{\mathbf{z}_i}{\|\mathbf{z}_i\|_2} \text{ and } \hat{\mathbf{w}}_i^T = \frac{\mathbf{w}_i}{\|\mathbf{w}_i\|_2} \quad \mathcal{L}_{\text{NormFace}}(\mathbf{Z}, \mathbf{W}) = \frac{1}{M} \sum_{i=1}^M -\log \left(\frac{e^{\frac{\mathbf{u}_i^T \hat{\mathbf{w}}_{y_i}}{\tau}}}{\sum_{j=1}^K e^{\frac{\mathbf{u}_i^T \hat{\mathbf{w}}_j}{\tau}}} \right)$$

Author's assess that **NormFace CE** can be seen as a contrastive reformulation of classification, in which the positives sample set has cardinality 1 (the correct prototype).

Problem: contrastive learning requires a large number on negative samples, which here is limited to $K-1$ (the number of classes that are not the correct one).

Proposal: within a batch, contrast the positive the true prototype against all elements in the batch

$$\mathcal{L}_{\text{NTCE}}(\mathbf{Z}, \mathbf{W}) = \frac{1}{M} \sum_{i=1}^M -\log \left(\frac{e^{\frac{\hat{\mathbf{w}}_{y_i} \mathbf{u}_i^T}{\tau}}}{\sum_{j=1}^M e^{\frac{\hat{\mathbf{w}}_{y_i} \mathbf{u}_j^T}{\tau}}} \right)$$

NTCE: Normalized Temperature-scaled Cross Entropy

Revisiting Cross Entropy 2/2

But what happens if the batch contains other elements of the same class of the current input?

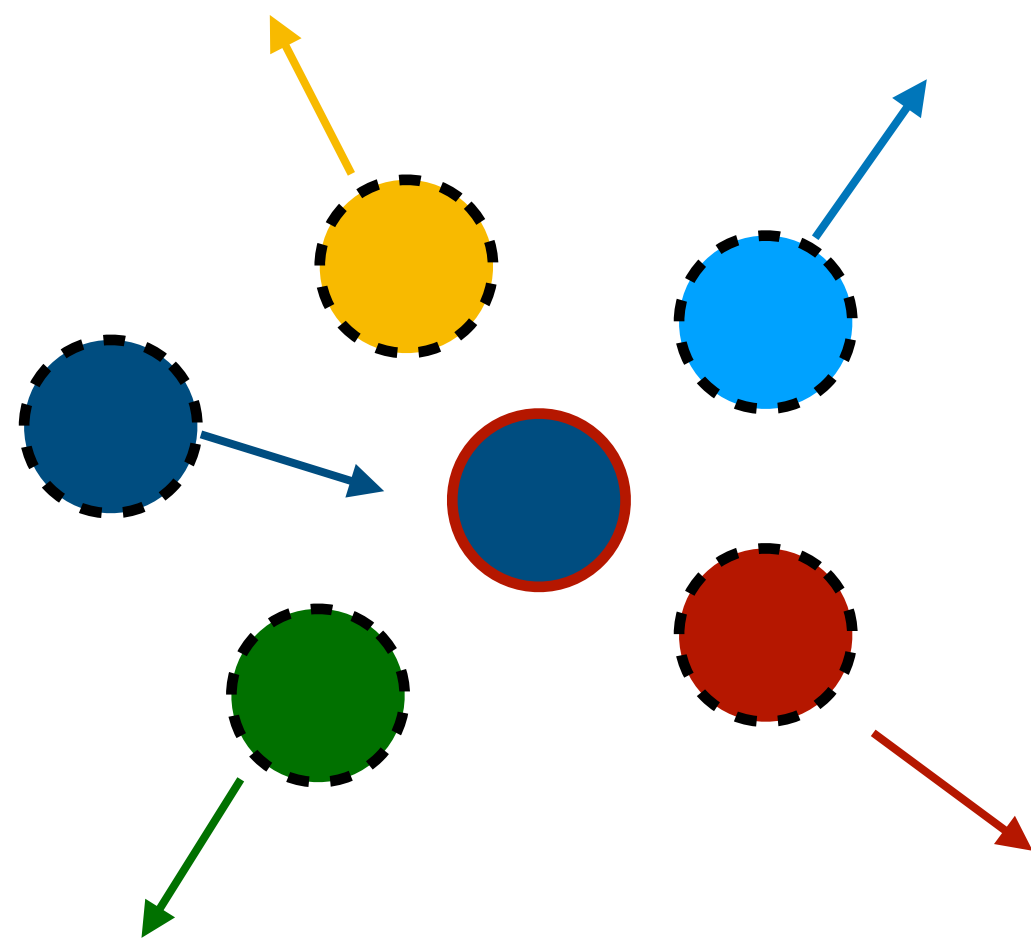
NTCE explicitly says to maximize the distance between elements of the same class!

Solution: NONL

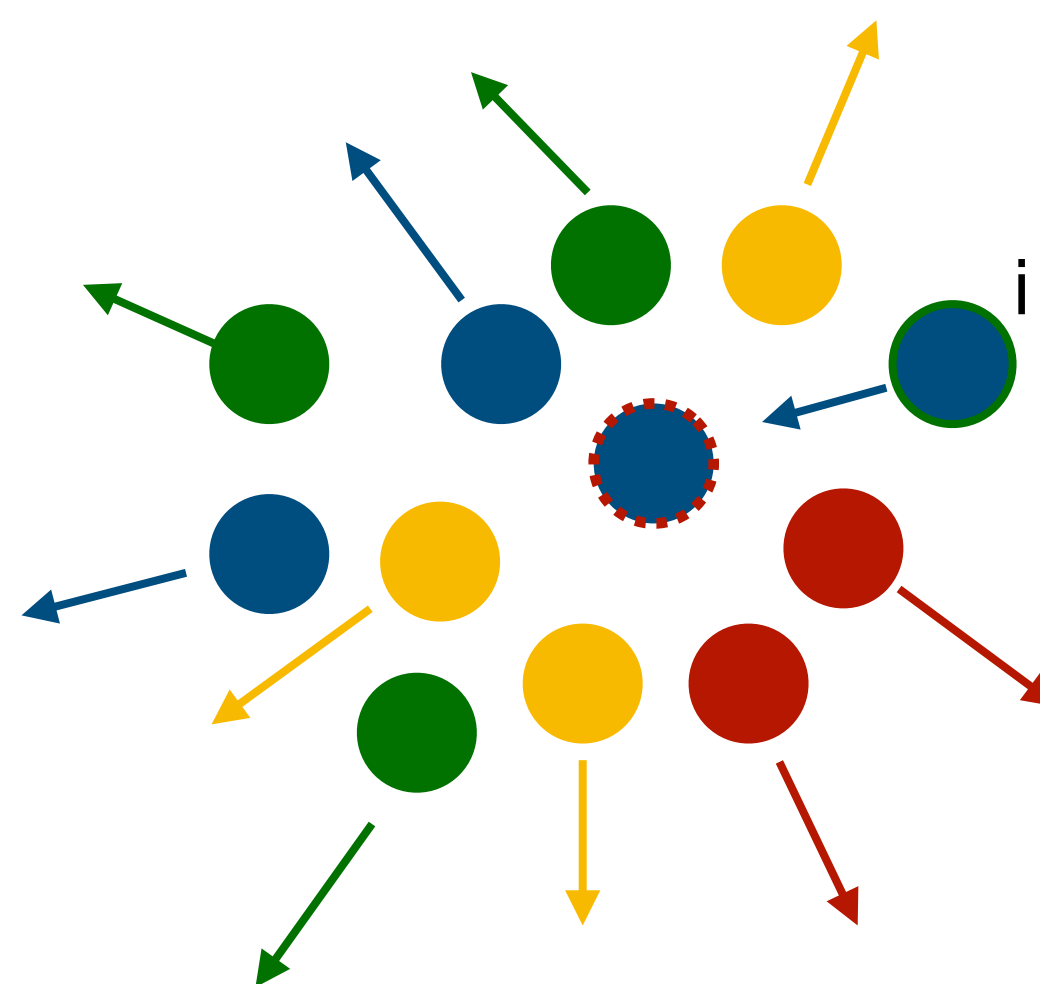
$$\mathcal{L}_{NONL}(\mathbf{Z}, \mathbf{W}) = \frac{1}{M} \sum_{i=1}^M -\log \left(\frac{e^{\frac{\hat{\mathbf{w}}_{y_i} \mathbf{u}_i^T}{\tau}}}{\sum_{j=1, j \neq C(i)}^M e^{\frac{\hat{\mathbf{w}}_{y_i} \mathbf{u}_j}{\tau}}} \right)$$

NONL: Negatives Only Normalization Loss

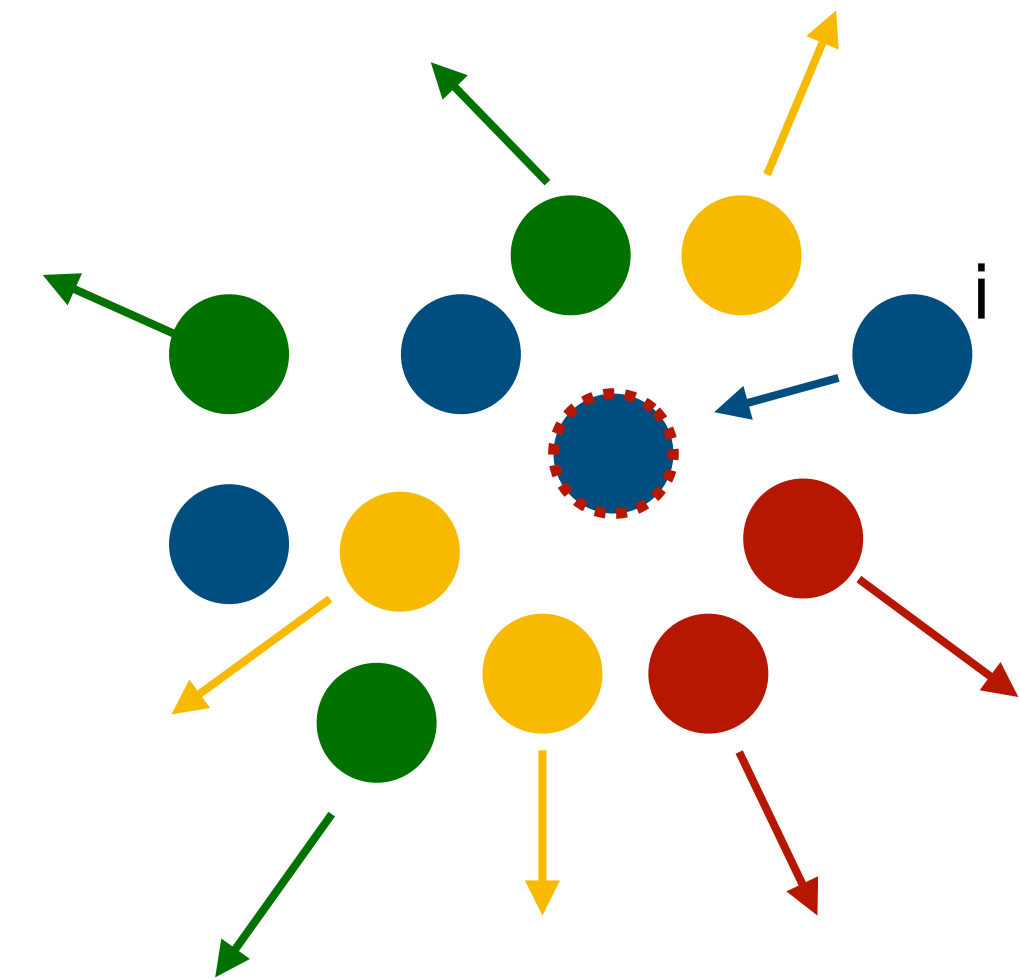
dotted-border: classes prototypes
continuous-border: samples
red border: anchor
i: current sample



NormFace: contrastive wrt to class prototypes

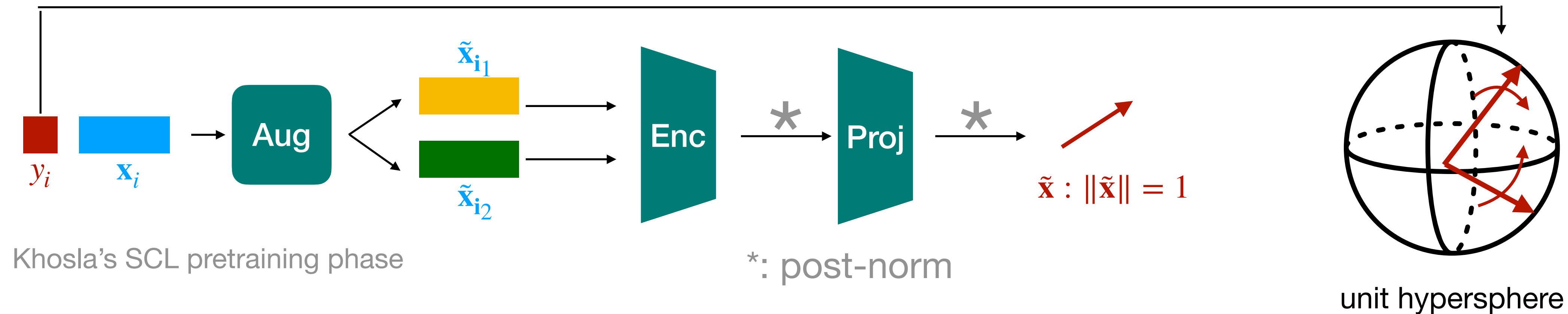


NTCE: prototype anchor pushing apart all elements within batch



NONL: avoid pushing apart same class samples

Revisiting SCL



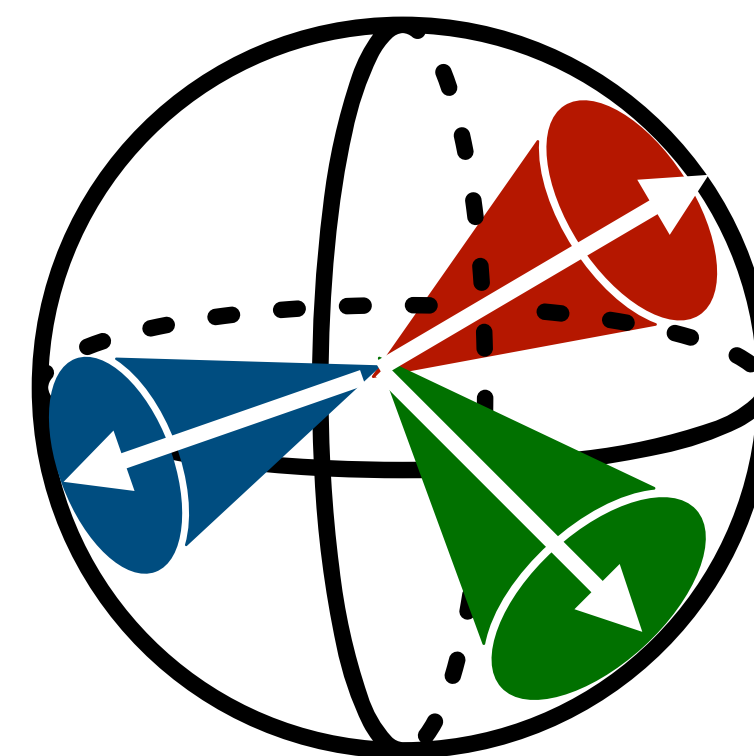
Authors claim: Khosla's posterior linear probing phase forces the classifier to work over a not normalized vector space, **discarding the geometry** achieved until that moment.

Solution: keeping the projection head, **instead of retraining** a new classifier for posterior tasks they build it by **using as weights the prototypes** (the mean classes of the representations on the hypersphere).

They call this: **Fixed Prototypes**

This is nice but:

Is it true that SCL guarantees NC?
Yes by Theorem 4.2



White arrows represents the class means, for each “cone” produced by standard SCL

Classification Performances

Loss	CIFAR-10	CIFAR-100	ImageNet-100	ImageNet-1K
CE	94.6	72.1	84.4	75.4
ETF + DR	94.4	72.1	84.5	75.4
NORMFACE	94.8	72.4	84.4	76.4
NTCE (ours)	94.7	72.9	84.7	76.7
NONL (ours)	94.9	73.6	84.9	76.5

Classifier Learning Performances - accuracy

(a) Normalized CE outperforms, or matches, “standard” **CE** over all considered datasets

(b) Author’s **revisited CE losses** outperform **Normalized CE**

(b.1) The **NTCE Loss** (due to the Contrastive Interpretation) clearly shows the limit of the contrastive approach on few negatives. Despite the anchor “pushes” away same class instances it performs better than **CE**.

(b.2) NONL, by removing from the negatives samples same class instances, shows better performances over all datasets except for ImageNet-1K

Method	Loss	Passes	CIFAR-10	CIFAR-100	IN-100	IN-1K
LP	SCL	$T \times N$	95.0	73.9	84.8	75.1
NLP	SCL	$T \times N$	94.9	73.6	84.8	75.1
FP (ours)	SCL	N	95.0	73.9	86.8	75.1

SCL Performances - accuracy

(a) Normalized LP is actually less effective wrt to Khosla’s original pipeline (only matches ImageNet-1K)

(b) Fixed Prototypes matches **LP**, outperforming it only over ImageNet-100. But we should consider that the Fixed Prototypes Classifier has been obtained “for free” by just **collecting centroids (prototypes)** after the projection module (**no need of a new training phase** as Khosla’s)

Evaluating Neural Collapse

Learning Family	Method	Effective Rank			Alignment		Information Theory Metrics	
		Intra ↓	Inter ↑	Weights ↑	Weight ↓	Instance ↓	MIR ↑	HDR ↓
CLASSIFIER LEARNING	CE	22.5 / 96.4	8.6 / 57.1	8.9 / 89.7	0.59 / 0.83	0.69 / 1.05	0.98 / 0.97	0.03 / 0.13
	ETF + DR	9.00 / 18.4	8.90 / 94.8	9.00 / 99.00	0.58 / 0.59	0.59 / 0.61	0.98 / 1.00	0.02 / 0.11
	NormFace	10.5 / 13.6	9.0 / 96.2	9.0 / 96.1	0.12 / 0.01	0.14 / 0.06	0.95 / 1.00	0.04 / 0.30
	NTCE	9.0 / 12.6	9.0 / 99.0	8.9 / 98.9	0.08 / 0.01	0.10 / 0.05	0.96 / 1.00	0.05 / 0.30
	NONL	4.0 / 11.4	9.0 / 99.0	9.0 / 99.0	0.11 / 0.01	0.16 / 0.06	0.95 / 1.00	0.05 / 0.30
CONTRASTIVE LEARNING	SCL (w probing)	4.5 / 7.5	9.0 / 66.7	8.3 / 77.8	0.99 / 1.03	0.10 / 0.34	0.99 / 0.95	0.07 / 0.11
	SCL (w/o probing)	4.5 / 7.5	9.0 / 66.7	9.0 / 66.5	0.00 / 0.00	0.10 / 0.34	1.00 / 0.87	0.09 / 0.14

(a) intra-class er (NC1) results shows that **SCL** performs better on larger class settings but **NONL** is a better choice in few-classes problems

(b) inter-class er (NC2/3) shows optimal results in **NTCE** and **NONL** for both few classes and more classes, this is also true for **SCL** configuration in few-classes problem, but definitely not optimal for more classes

(c) NONL achieves optimal results wrt to **Weights ETF (NC2-Weights)**, the same as **ETF+DR**.

(d) SCL w/o probing achieves optimal results wrt to **Weights alignments (NC3)**, while probing destroys it.

Conclusions

Standard Cross Entropy and SCL are not enough to reach NC

NONL shows optimal results in reaching NC almost anytime

Fixed Prototypes SCL without linear probing achieve optimal results for free

Thanks!

References

[1] Koromilas, Panagiotis, et al. "Neural Collapse by Design: Learning Class Prototypes on the Hypersphere." *arXiv preprint arXiv:2605.20302* (2026).

[2] Khosla, Prannay, et al. "Supervised contrastive learning." *Advances in neural information processing systems* 33 (2020): 18661-18673.

* Presentation produced via Keynote app (MacOS).

* Gemini has been used in order to arrange the single instance cross-entropy Latex equation in slide 3.